

1 Last year homework

Exercise

The exercise is a part of the book by Baltagi¹ (p. 175-176).

Baltagi and Levine (1986) estimated a dynamic panel model for cigarettes based on panel data from 46 American states. This data, updated from 1963-1992, is available on the Wiley website as the file `cigar.txt`. Let's analyse a static version of the model, that is

$$\ln sales_{i,t} = \alpha + \beta_1 \ln price_{i,t} + \beta_2 \ln ndi_{i,t} + \beta_3 \ln pimin_{i,t} + \mu_i + \lambda_t + \varepsilon_{i,t} \quad (1)$$

where the subscript i denotes the i th state ($i = 1, \dots, 46$) and the subscript t denotes the t th year ($t = 1, \dots, 32$). Here, $sales_{i,t}$ represents real per capita sales of cigarettes by persons of smoking age (14 years and older), measured in packs of cigarettes per head; $P_{i,t}$ is the average retail price of a pack of cigarettes measured in real terms; $ndi_{i,t}$ is real per capita disposable income; and $pimin_{i,t}$ denotes the minimum real price of cigarettes in any neighbouring state. This last variable is a proxy for the casual smuggling effect across state borders. It acts as a substitute price attracting consumers from high-tax states such as Massachusetts, where the tax is 26 cents per pack, to cross over to New Hampshire, where tax is only 12 cents per pack. The disturbance term is specified as a two-way error component model

$$\mu_i + \lambda_t + \varepsilon_{i,t} \quad (2)$$

where μ_i denotes a state-specific effect and λ_t denotes a year-specific effect. The time-period effects (the λ_t) are assumed to be fixed parameters to be estimated as coefficients of time dummies for each year in the sample. This can be justified given the numerous policy interventions as well

as health warnings, and Surgeon General's reports, for example:

1. the imposition of warning labels by the Federal Trade Commission effective January 1965;
2. the application of the Fairness Doctrine Act to cigarette advertising in June 1967, which subsidised anti-smoking messages from 1968 to 1970;
3. the Congressional ban on broadcast advertising of cigarettes effective January 1971.

The μ_i are state-specific effects which can represent any state-specific characteristic, including the following:

1. States with Indian reservations, such as Montana, New Mexico and Arizona, are among the biggest losers of tax revenues from non-Indians purchasing tax-exempt cigarettes from the reservations.
2. Florida, Texas, Washington and Georgia are among the biggest losers of tax revenues due to the purchase of cigarettes from tax-exempt military bases in these states.
3. Utah, where a high percentage of the population follows Mormonism (a religion which forbids smoking), had per capita cigarette sale of 55 packs in 1998, a little under half the national average of 113 packs.
4. Nevada, which is a highly touristic state, had per capita cigarette sales of 142 packs in 1988, 29 more packs than the national average.

These state-specific effects may assumed to be fixed, in which case one includes state dummy variables in equation (1).

¹B.H. Baltagi (2013), *Econometric Analysis of Panel Data, Fifth Edition*, Wiley.

- a) What are economic reasons to analyse demand for cigarettes?
- b) Which estimator out of fixed-effects and random-effects should we choose on the ground of theory?
- c) Which model out of FE, RE, and POLS should be selected using appropriate tests?
- d) The quality publication table on the next page presents estimates of POLS, fixed-effects and random-effects models. Interpret estimates of price, income and *pimin* of the fixed-effects model.
- e) How to check whether year-specific dummies should be removed out of the model? Write down the null hypothesis.
- f) What are variables the model could be extended with? Explain your proposal.

Hausman Test

```
data: lnC ~ lnP + lnPn + lnY + as.factor(year)
chisq = 1.7158, df = 32, p-value = 1
alternative hypothesis: one model is inconsistent
```

```
> #individual effects for random effects?
> plmtest(POLS, type=c("bp"))
```

Lagrange Multiplier Test - (Breusch-Pagan) for balanced panels

```
data: lnC ~ lnP + lnPn + lnY + as.factor(year)
chisq = 12738, df = 1, p-value < 2.2e-16
alternative hypothesis: significant effects
```

```
> #individual effects for fixed effects?
> pFtest(fixed, POLS)
```

F test for individual effects

```
data: lnC ~ lnP + lnPn + lnY + as.factor(year)
F = 125.73, df1 = 45, df2 = 1302, p-value < 2.2e-16
alternative hypothesis: significant effects
```

Dependent variable:			
	lnSALES		
	(1)	(2)	(3)
lnPrice	-1.277***	-1.023***	-1.026***
lnPimin	0.160**	-0.117*	-0.110*
lnNDI	0.569***	0.520***	0.522***
as.factor(year)64	-0.047	-0.041**	-0.042**
as.factor(year)65	-0.085*	-0.072***	-0.072***
as.factor(year)66	-0.103**	-0.081***	-0.082***
as.factor(year)67	-0.139***	-0.106***	-0.107***
as.factor(year)68	-0.169***	-0.119***	-0.121***
as.factor(year)69	-0.246***	-0.182***	-0.184***
as.factor(year)70	-0.284***	-0.203***	-0.206***
as.factor(year)71	-0.293***	-0.198***	-0.201***
as.factor(year)72	-0.304***	-0.199***	-0.202***
as.factor(year)73	-0.425***	-0.298***	-0.303***
as.factor(year)74	-0.541***	-0.384***	-0.389***
as.factor(year)75	-0.611***	-0.425***	-0.431***
as.factor(year)76	-0.623***	-0.416***	-0.422***
as.factor(year)77	-0.725***	-0.496***	-0.503***
as.factor(year)78	-0.763***	-0.506***	-0.514***
as.factor(year)79	-0.925***	-0.632***	-0.641***
as.factor(year)80	-1.070***	-0.740***	-0.749***
as.factor(year)81	-1.192***	-0.831***	-0.841***
as.factor(year)82	-1.200***	-0.817***	-0.828***
as.factor(year)83	-1.140***	-0.745***	-0.757***
as.factor(year)84	-1.142***	-0.728***	-0.741***
as.factor(year)85	-1.164***	-0.737***	-0.751***
as.factor(year)86	-1.166***	-0.727***	-0.741***
as.factor(year)87	-1.197***	-0.747***	-0.761***
as.factor(year)88	-1.236***	-0.773***	-0.787***
as.factor(year)89	-1.275***	-0.795***	-0.810***
as.factor(year)90	-1.317***	-0.819***	-0.835***
as.factor(year)91	-1.346***	-0.838***	-0.854***
as.factor(year)92	-1.295***	-0.771***	-0.788***
Constant	5.679***		5.785***
Observations	1,380	1,380	1,380
R2	0.443	0.678	0.672
Adjusted R2	0.430	0.659	0.664
F Statistic	33.531*** (df = 32; 1347)	85.646*** (df = 32; 1302)	2,763.110***
Note: *p<0.05; **p<0.01; ***p<0.001			